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BACK SUPPORT ON SEAT

Abstract

A device may include a support strip disposed adjacent to a lower position of a seatback frame of a passenger seat. A device may include a first cable coupled on a first end to a first strip end of the support strip. A device may include a first cable adjustor coupled to a second end of the first cable, wherein an actuation of the first cable causes a movement of the support strip from a back position to a forward position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims the benefit under 35 U.S.C. § 119 (e) of Indian Provisional Application Ser. No. 20/241,1015823, filed Mar. 6, 2024, naming Krishnateja Pillarisetty as inventor, which is incorporated herein by reference in the entirety.

BACKGROUND

[0002] Aircraft passengers often suffer back pain during flight, particularly during long flights where the passenger needs to sit down for extended periods of time. Therefore, there is a need for a system that supports the back of the passengers as they sit in the passenger seat.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a system including: a support strip disposed adjacent to a lower position of a seatback frame of a passenger seat; a first cable coupled on a first end to a first strip end of the support strip; a first cable adjustor coupled to a second end of the first cable, wherein an actuation of the first cable causes a movement of the support strip from a back position to a forward position.

[0004] In some aspects, the techniques described herein relate to a system, wherein the first cable includes a Bowden cable, the Bowden cable including: an inner cable coupled to the cable adjustor and the first end of the support strip; and an outer cable.

[0005] In some aspects, the techniques described herein relate to a system, wherein the first cable adjustor includes a first knob, wherein the first knob includes an axle coupled to the inner cable, wherein a rotation of the first knob causes the inner cable to wind onto the axle and causes a movement of the support strip from a back position to a forward position.

[0006] In some aspects, the techniques described herein relate to a system, wherein the first cable adjustor is integrated into a first armrest of the passenger seat.

[0007] In some aspects, the techniques described herein relate to a system, wherein the first cable adjustor is integrated into a first armrest of the passenger seat.

[0008] In some aspects, the techniques described herein relate to a system, further including the first armrest and the seatback

[0009] In some aspects, the techniques described herein relate to a system, wherein the first cable adjustor includes at least one of a lever or pull handle.

[0010] In some aspects, the techniques described herein relate to a system, further including the passenger seat.

[0011] In some aspects, the techniques described herein relate to a system, further including: a second cable including a first end, the first end of the second cable coupled to a second end of the support strip; a second cable adjustor coupled to a second end of the second cable, wherein an actuation of the second cable causes a movement of the support strip from a back position to a forward position.

[0012] In some aspects, the techniques described herein relate to a system, wherein the first cable includes a first Bowden cable, the first Bowden cable including: an inner cable coupled to the first cable adjustor and the first end of the support strip; and an outer cable, wherein the second cable includes a second Bowden cable, the second Bowden cable including: an inner cable coupled to the second cable adjustor and the second end of the support strip; and an outer cable.

[0013] In some aspects, the techniques described herein relate to a passenger seat including a seat pan; a seat back frame; a first armrest; a support strip disposed adjacent to a lower position of a seatback frame of the passenger seat; a first cable coupled on a first end to the support strip; a first cable adjustor integrated into the first armrest and coupled to a second end of the first cable, wherein an actuation of the cable causes a movement of the support strip from a back position to a forward position.

[0014] In some aspects, the techniques described herein relate to a passenger seat, further including: a second cable including a first end, the first end of the second cable coupled on a first

end to a second end of the support strip; a second cable adjustor coupled to a second end of the second cable, wherein an actuation of the second cable causes a movement of the support strip from a back position to a forward position.

[0015] In some aspects, the techniques described herein relate to a passenger seat, wherein the first cable includes a first Bowden cable, the first Bowden cable including: an inner cable coupled to the first cable adjustor and the first end of the support strip; and an outer cable, wherein the second cable includes a second Bowden cable, the second Bowden cable including: an inner cable coupled to the second cable adjustor and the second end of the support strip; and an outer cable.

[0016] In some aspects, the techniques described herein relate to a passenger seat, wherein the first cable adjustor includes a first knob, wherein the knob includes an axle coupled to the inner cable, wherein a rotation of the first knob causes the inner cable to wind onto the axle and causes a movement of the support strip from a back position to a forward position, wherein the second cable adjustor includes a second knob, wherein the second knob includes an axle coupled to the inner cable, wherein the rotation of the second knob causes the inner cable to wind onto the axle and causes a movement of the support strip from a back position to a forward position.

[0017] In some aspects, the techniques described herein relate to a method including: sitting in a passenger seat; rotating a knob integrated into an armrest of the passenger seat; and moving a support strip of the passenger seat from a back position to a forward position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The detailed description is described with reference to the accompanying figures. The use of the same reference numbers in different instances in the description and the figures may indicate similar or identical items. Various embodiments or examples (“examples”) of the present disclosure are disclosed in the following detailed description and the accompanying drawings. The drawings are not necessarily to scale. In general, operations of disclosed processes may be performed in an arbitrary order, unless otherwise provided in the claims. In the drawings:

[0019] FIG. 1 illustrates a perspective front view of a set of passenger seats.

[0020] FIG. 2 illustrates a perspective front view of a seatback frame and armrests of a passenger seat.

[0021] FIG. 3 illustrates a perspective view of a back support mechanism.

[0022] FIG. 4 illustrates a perspective view of a back support mechanism, with cable adjustors shown in detail.

[0023] FIG. 5. illustrates a close-up view of a cable adjustor.

[0024] FIG. 6 illustrates a pan view of a passenger seat that includes a back support mechanism.

[0025] FIG. 7 illustrates closeup pan views of a passenger seat with a support strip configured in a back position and a forward position.

[0026] FIG. 8 illustrates a side view of portions of a seatback frame, a spreader, armrests, and the back support mechanism. FIG. 8 further includes in an inset the armrest and a cable and knob of the back support mechanism integrated into the armrest.

[0027] FIG. 9 is a flow chart illustrating a method for using a passenger seat equipped with a back support mechanism.

DETAILED DESCRIPTION

[0028] Before explaining one or more embodiments of the disclosure in detail, it is to be understood that the embodiments are not limited in their application to the details of construction and the arrangement of the components or steps or methodologies set forth in the following description or illustrated in the drawings. In the following detailed description of embodiments, numerous specific details may be set forth in order to provide a more thorough understanding of the

disclosure. However, it will be apparent to one of ordinary skill in the art having the benefit of the instant disclosure that the embodiments disclosed herein may be practiced without some of these specific details. In other instances, well-known features may not be described in detail to avoid unnecessarily complicating the instant disclosure.

[0029] As used herein a letter following a reference numeral is intended to reference an embodiment of the feature or element that may be similar, but not necessarily identical, to a previously described element or feature bearing the same reference numeral (e.g., **1**, **1a**, **1b**). Such shorthand notations are used for purposes of convenience only and should not be construed to limit the disclosure in any way unless expressly stated to the contrary.

[0030] Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by anyone of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0031] In addition, use of “a” or “an” may be employed to describe elements and components of embodiments disclosed herein. This is done merely for convenience and “a” and “an” are intended to include “one” or “at least one,” and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0032] Finally, as used herein any reference to “one embodiment” or “some embodiments” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment disclosed herein. The appearances of the phrase “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, and embodiments may include one or more of the features expressly described or inherently present herein, or any combination of sub-combination of two or more such features, along with any other features which may not necessarily be expressly described or inherently present in the instant disclosure.

[0033] Broadly, embodiments of the inventive concepts disclosed herein are directed to a method and a system, which may include a support strip disposed adjacent to a lower position of a seatback frame of a passenger seat and at least one cable coupled to one or more ends of the support strip. The system may further include one or more cable adjustors (e.g., a knob) integrated into one or more armrests of the passenger seat. When sitting in the passenger seat, a passenger can actuate the adjustor (e.g., turn the knob) to adjust the position (e.g., forward or backward) of the support strip relative to the seatback frame. In this manner, the passenger can adjust the support strip to a comfortable position.

[0034] Referring now to FIGS. **1-9**, embodiments of the system **100** according to the concepts disclosed herein are depicted. The system **100** may be implemented into any suitable system (e.g., chair, seat, bench) and may be further implemented into any suitable vehicle (e.g., aircraft, bus, automobile). For example, the system **100** may include a passenger seat **104a-c** or one or more components of the passenger seat **104a-c**, as shown in FIG. **1**. For instance, the system **100** may include a seatback frame **108a-c**, a seat pan **112a-c**; and one or more armrests **116a-d**. The system **100** may further include a support strip **120** that supports the back of a passenger. The support strip **120** may be configured in any supportive structure or shape including but not limited to, a strap.

[0035] In a close-up view of the seatback frame **108** and the armrests **116a-b**, the support strip is shown positioned adjacent to a lower section **204** (e.g., a lumbar section) of the seatback frame **108**, as shown in FIG. **2**. In embodiments, the system **100** further includes a first cable **208** coupled to the support strip **120**. The first cable **208** may be at least partially integrated within a first armrest **116a**. The system may also include a second cable **212** coupled to the support strip **120**. The second cable may be at least partially integrated within a second armrest **116b**. The system may therefore include a support strip **120** that is coupled at least to a first cable **208**, or to a first cable **208** and a second cable **212**.

[0036] A closeup and isolated view of the support strip **120**, first cable **208**, and second cable (e.g.,

collectively, a back support mechanism **300**) is shown in FIG. 3. The first cable **208** is coupled on a first end **304** to a first strip end of the support strip **120**. the second cable **212** is coupled on a first end **312** to a second strip end **316** of the support strip **120**. The first cable **208** is coupled on a second end **320** to a first adjustor **324**. The second cable **212** is coupled on a second end **328** to a second adjustor **332**. The first adjustor **324** and the second adjustor **332** are configured so that when the first adjustor **324** and/or the second adjustor **332** are activated, the support strip **120** moves (e.g., translates) from a back position **340** to a forward position **344** or vice-versa.

[0037] The first cable **208** and/or the second cable **212** may include any type of cable or cable system including but not limited to a Bowden cable. For example, the first cable **208** and/or the second cable **212** may include a Bowden cable comprising an outer cable **404** (e.g., casing) and an inner cable **408**, as illustrated in FIG. 4. In use, the first adjustor **324** and/or the second adjustor **332** is coupled directly to the inner cable **408** and can slid the inner cable **408** relative to the outer cable **404**. The sliding of the inner cable **408** then causes the movement or translation of the support strip **120** from the back position to the forward position. The first cable **208** and/or the second cable **212** may include other componentry common to Bowden cables including but not limited to a ferrule, a stop sleeve, an adjusting screw, a towing eye, and an anchor (e.g., nipple).

[0038] The first adjustor **324** and/or the second adjustor **332** may be configured as any type of cable adjusting mechanism, such as cable adjusting mechanisms that can couple to the inner cable **408** and cause the sliding of the inner cable **408** relative to the outer cable **404**. Types of cable adjusting mechanisms include but are not limited to lever, pull handles, and twisting mechanisms. For example, the first adjustor **324** and/or the second adjustor **332** may include a knob **412** coupled to the inner cable **408** that can twist or turn so that the inner cable **408** wraps or winds at least partially around an axle. The knob **412** may include one or more protrusions or recesses **504** that aid the passenger in gripping the knob **412** in order to twist the knob clockwise **508** or counterclockwise **512**, as illustrated in FIG. 5.

[0039] An example of the positioning of the support strip **120**, first cable **208**, second cable **212**, and the knob **412a-b** on the passenger seat **104** are shown in FIG. 6. The support strip support strip **120** is disposed adjacent to a front surface **604** of the seatback frame **108**. The seatback frame **108** is shown coupled to the seat pan **112** via a linker **608**. The first cable **208** and/or the second cable **212** may feed into, or be integrated into, their respective armrest **116a-b**. The knobs **412** are positioned along the armrest. For example, the knobs **412a-b** may be positioned at a point along the armrests **116a-b** near where the fingers of the passenger are predicted to be placed. The support strip **120** may be designed to follow the contour of the seatback frame **108** or the back of a passenger.

[0040] The movement of the support strip **120** from a back position **704** to a forward position **708** is illustrated in FIG. 7. In the back position **704**, the support strip **120** is positioned close to the front surface **604** of the seatback frame **108**. Upon movement **712** (e.g., translation) to the forward position **708**, the support strip **120** is moved to a position forward (e.g., aft) relative to the front surface **604** of the seatback frame **108**. The movement of the support strip **120** (e.g., when moving from the back position **704** to the forward position **708**) may be approximately or substantially one centimeter, may be approximately or substantially two centimeters, may be approximately or substantially 2.5 centimeters, may be approximately or substantially three centimeters, may be approximately or substantially four centimeters, or may be approximately or substantially five or more centimeters. For example, the movement of the support strip **120** when moving from the back position **704** to the forward position **708** may be approximately or substantially 2.55 centimeters (e.g., on inch).

[0041] A side view of an armrest **116** (adjacent to the seatback frame **108** and linked to a spreader **804**) is illustrated in FIG. 8. The position support strip **120** is shown for clarity. In embodiments, the first cable **208** is integrated into the armrest **116** via one or more apertures **808** that lead the first cable to a cavity **810** inside the armrest **116**, as shown in inset **812**. The first cable **208** then

connects to the knob **412**.

[0042] In embodiments, the system **100** includes the support strip **120**, the first cable **208**, and the first cable adjuster **324** (e.g., such as the knob). In embodiments, the system **100** may further include the second cable **212**, the second cable **212** and the second cable adjuster **332**. In embodiments, the system **100** may further include the seatback frame **108** and the armrest **116**. In embodiments, the system may further include the passenger seat **104**.

[0043] A method **900** for utilizing the system **100** is also disclosed. The method may be utilized by a passenger adjusting the system **100** in order to sit more comfortably in the passenger seat. In embodiments, the method **900** includes a step **910** of sitting in the passenger seat **104** (e.g., a passenger seat **104** that includes the system **100**). In embodiments, the method **900** includes a step **920** of rotating the knob **412** integrated into the armrest **116** of the passenger seat **104**. For example, the passenger may rotate one knob **412**, or both knobs **412** (e.g., one on each armrest **116a-b**). In embodiments, the method includes a step **930** of moving the support strip **120** from a back position **704** to the forward position **708**.

[0044] It is to be understood that embodiments of the methods disclosed herein may include one or more of the steps described herein. Further, such steps may be carried out in any desired order and two or more of the steps may be carried out simultaneously with one another. Two or more of the steps disclosed herein may be combined in a single step, and in some embodiments, one or more of the steps may be carried out as two or more sub-steps. Further, other steps or sub-steps may be carried in addition to, or as substitutes to one or more of the steps disclosed herein.

[0045] Although inventive concepts have been described with reference to the embodiments illustrated in the attached drawing figures, equivalents may be employed and substitutions made herein without departing from the scope of the claims. Components illustrated and described herein are merely examples of a system/device and components that may be used to implement embodiments of the inventive concepts and may be replaced with other devices and components without departing from the scope of the claims. Furthermore, any dimensions, degrees, and/or numerical ranges provided herein are to be understood as non-limiting examples unless otherwise specified in the claims.

Claims

1. A system comprising: a support strip disposed adjacent to a lower position of a seatback frame of a passenger seat; a first cable coupled on a first end to a first strip end of the support strip; and a first cable adjuster coupled to a second end of the first cable, wherein an actuation of the first cable causes a movement of the support strip from a back position to a forward position.
2. The system of claim 1, wherein the first cable comprises a Bowden cable, the Bowden cable comprising: an inner cable coupled to the first cable adjuster and the first end of the support strip; and an outer cable.
3. The system of claim 2, wherein the first cable adjuster comprises a first knob, wherein the first knob comprises an axle coupled to the inner cable, wherein a rotation of the first knob causes the inner cable to wind onto the axle and causes a movement of the support strip from a back position to a forward position.
4. The system of claim 3, wherein the first cable adjuster is integrated into a first armrest of the passenger seat.
5. The system of claim 2, further including a first armrest and the seatback frame
6. The system of claim 5, wherein the first cable adjuster is integrated into the first armrest of the passenger seat.
7. The system of claim 2, further including the passenger seat.
8. The system of claim 1, wherein the first cable adjuster comprises at least one of a lever or pull handle.

9. The system of claim 1, further comprising: a second cable comprising a first end, the first end of the second cable coupled to a second end of the support strip; a second cable adjuster coupled to a second end of the second cable, wherein an actuation of the second cable causes a movement of the support strip from a back position to a forward position.

10. The system of claim 9, wherein the first cable comprises a first Bowden cable, the first Bowden cable comprising: an inner cable coupled to the first cable adjuster and the first end of the support strip; and an outer cable, wherein the second cable comprises a second Bowden cable, the second Bowden cable comprising: an inner cable coupled to the second cable adjuster and the second end of the support strip; and an outer cable.

11. A passenger seat comprising a seat pan; a seat back frame; a first armrest; a support strip disposed adjacent to a lower position of a seatback frame of the passenger seat; a first cable coupled on a first end to the support strip; a first cable adjuster integrated into the first armrest and coupled to a second end of the first cable, wherein an actuation of the first cable adjuster causes a movement of the support strip from a back position to a forward position.

12. The passenger seat of claim 11, further comprising: a second cable comprising a first end, the first end of the second cable coupled on a first end to a second end of the support strip; a second cable adjuster coupled to a second end of the second cable, wherein an actuation of the second cable causes a movement of the support strip from a back position to a forward position.

13. The passenger seat of claim 12, wherein the first cable comprises a first Bowden cable, the first Bowden cable comprising: an inner cable coupled to the first cable adjuster and the first end of the support strip; and an outer cable, wherein the second cable comprises a second Bowden cable, the second Bowden cable comprising: an inner cable coupled to the second cable adjuster and the second end of the support strip; and an outer cable.

14. The passenger seat of claim 13, wherein the first cable adjuster comprises a first knob, wherein the first knob comprises an axle coupled to the inner cable, wherein a rotation of the first knob causes the inner cable to wind onto the axle and causes a movement of the support strip from a back position to a forward position, wherein the second cable adjuster comprises a second knob, wherein the second knob comprises an axle coupled to the inner cable, wherein the rotation of the second knob causes the inner cable to wind onto the axle and causes a movement of the support strip from a back position to a forward position.

15. A method comprising: sitting in a passenger seat; rotating a knob integrated into an armrest of the passenger seat; and moving a support strip of the passenger seat from a back position to a forward position.
